```
class FatigueResistantCMA:
    def __init__(self, N, λ=4+4*N):
        self.λ = λ
        self.t = 0
        self.fatigue_counter = 0

    def step(self, population):
        self.t += 1

        # Детекция усталости
        p_success = success_rate(population)
        diversity = pop_std(population)

        if p_success < 0.1 or diversity < 0.05:
            self.fatigue_counter += 1
        else:
            self.fatigue_counter = 0

        # Антиусталостные меры
        if self.fatigue_counter > 3 or self.t % 100
== 0:
            self.inject_immigrants(0.15)
```

```
        self.active_cov_update()

    return self.next_generation()
```